

Justin (Chengyang) Zhang

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EDUCATION

University of California - Los Angeles

B.S. Cognitive Science (Specialization in Computing), B.A. Linguistics and CS

Los Angeles, CA
Expected June 2029

- Academics and Test Scores: 3.89 Cumulative GPA
- Activities and Honors: Dean's Honor List x2, Bruin Linguists Society, UCLA Radio
- Coursework: Algorithm and Complexity, NLP in Social Sciences, Intro to Computer Science I & II, Intro to Discrete Structures, Linear Algebra, Intro to Cognitive Science, Intro to Probability, Research Methods in Psychology, Language in Culture, Meaning and Communication, Intro to Linguistic Analysis

RESEARCH EXPERIENCE

Cambridge University, Prof. Weiwei Sun

Student Researcher

Remote
Jan 2026 - present

- Developed a philologist-inspired category-based optimization model for joint Middle Chinese-Old Mandarin reconstruction, replacing character-level prediction with category-level phonological inference and explicit constraints on the directionality of historical sound change.
- Modeled historical reconstruction through distinctive features, modern dialect reflexes, fanqie, rhyme-table structure, and cross-period sound-change constraints, with evaluation against expert philological reconstructions and Sino-Xenic evidence.
- Added 6 new historical time plane based on Wang Li's phonological system, reconstructing its inventory from available historical evidence, preparing submission to Computational Linguistics.

Peking University Century Jinyuan "Tengyun" Clinical Translation Project

Project Lead Developer

Beijing, China
Jul 2026 - present

- Developing a hybrid physics-informed machine learning system for hemodialysis adequacy prediction and prescription optimization to support nephrologists in clinical decision-making.
- Trained XGBoost and CatBoost models for spKt/V and HDF convective volume on 246,590 longitudinal sessions, using patient-level and temporal validation and achieving a held-out spKt/V MAE of 0.118.
- Built an interactive tool that recommends blood-flow-rate and treatment-duration combinations.

UCLA, Perceptual Processing and Computational (Zili) Lab

Project Lead

Los Angeles, CA
Oct 2025 - present

- Contributed to a psychophysics study of stereokinetic depth perception, implementing real-time hand-pose acquisition via SteamVR/OpenXR and Unity/C# experiment control with calibration, practice, and multi-trial sessions.
- Contributed to evaluating face-privacy perturbations across 272 participants, 4,254 face pairs, and five recognition models spanning identical twins, fraternal twins, and doppelgangers.
- Implementing DiffAM diffusion-based adversarial makeup for complementary privacy evaluation, including model fine-tuning, protected-image generation, cross-model transfer analysis, and crop-preserving preprocessing.

UCLA, Language Processing Lab, Prof. Jesse Harris

Research Assistant

Los Angeles, CA
Mar 2026 - present

- Investigating how semantic plausibility shapes real-time processing of filler-gap dependencies, comparing human processing patterns with predictions from language models.
- Developing a GPT-2/HuggingFace surprisal pipeline to extract token-level expectations at critical Verb, Object, and Adverb regions across controlled experimental conditions.
- Preparing and validating linguistic stimuli, aligning model predictions with experimental factors, and producing analysis-ready data for human-LLM comparisons of syntactic processing.

Peking University, School of Psychological and Cognitive Sciences - (Mao Lab)

Research Assistant/Intern

Beijing, China
Sep 2024 - Sep 2025

- Designed a high-sensitivity 3D Spatial Reproduction Paradigm and conducted 5 behavioral + VR experiments, revealing systematic interactions between temporal and spatial magnitude perception, with VR tasks built in Unity Quest 3.
- Conducted cross-linguistic gesture analysis examining how speakers of typologically distinct languages encode temporal and spatial magnitude through co-speech gesture, identifying divergences in axis orientation and gesture-speech synchrony.

UCLA, Undergraduate Research Week, Prof. Sam Cumming

Student Researcher

Los Angeles, CA
Jan 2026 - Apr 2026

- Constructed a 23M-character diachronic corpus and trained period-specific Word2Vec models with Orthogonal Procrustes alignment to quantify semantic shift via cosine distance z-scores.
- Engineered theoretically-motivated semantic axes via contrastive word-pair projection in Python (gensim, scikit-learn) to track pejorative, urban-coding, and cultural-identity trajectories across three historical periods.

ADDITIONAL

Awards: Finalist of The New York Times "Where We Are" Photo Essay Contest, Merit of John Locke Global Essay Contest, AP Scholar with Distinction, Silver Medal of iGEM Synthetic Biology Competition, USABO Bronze Medal

Skills: Microsoft Suite, OpenAI, Adobe Photoshop, R, SPSS, Qualtrics, Praat, C#, Python, C++, HTML, LingPy, PyTorch, TensorFlow, Javascript, Mandarin, Cantonese, Spanish.